

Tech & Teach

Digital Chip Design Program

RTL Design

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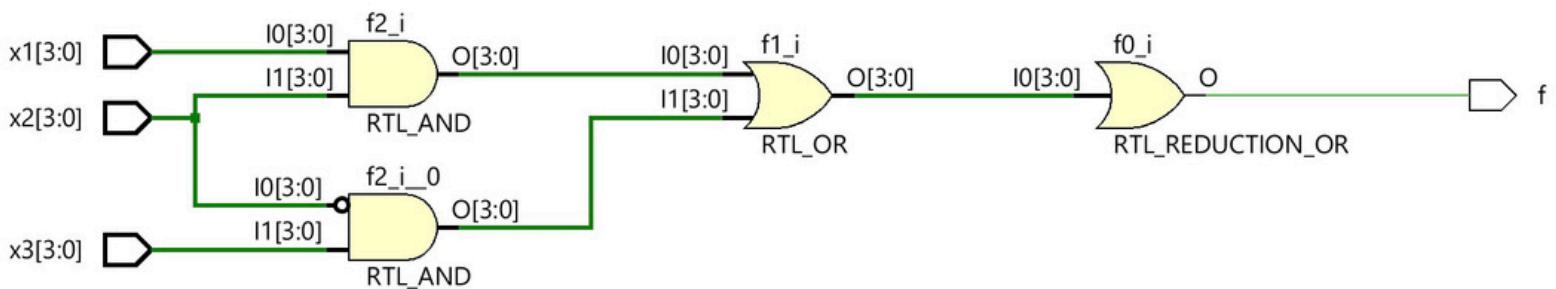
1/7/2026

Behavioral

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Task #3.1

```
module task3
#(parameter n=4)(
input logic [n-1:0] x1,x2,x3,
output logic f );
always@ (*) begin
    if((x1 & x2) | (~x2&x3))
        f=1;
    else
        f=0;
    end
endmodule
```

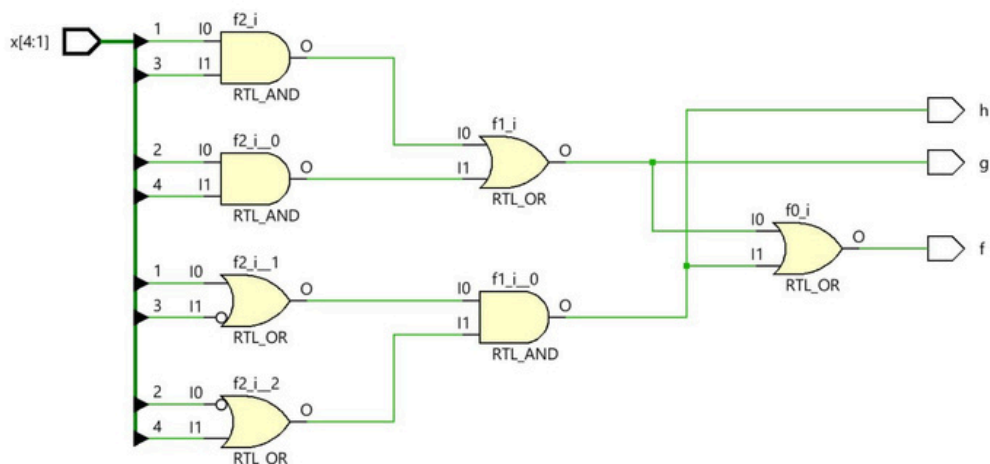
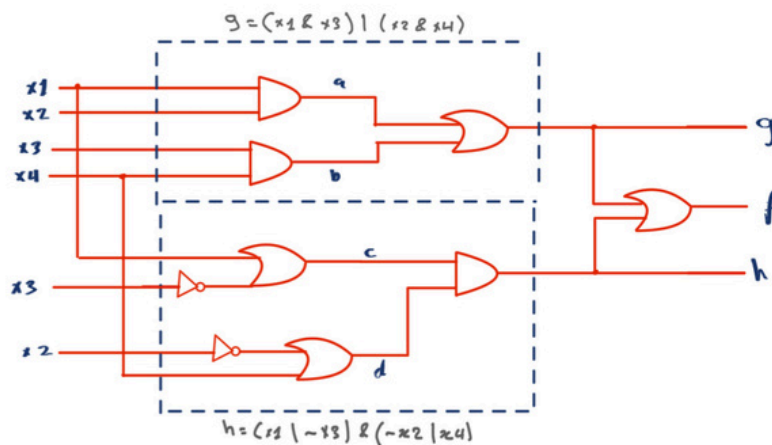


Task #3.2

```

module comb_circuit(f, g, h, x);
input [4:1] x;
output reg f, g, h;
always @(*) begin
  g = (x[1] & x[3]) | (x[2] & x[4]);
  h = (x[1] | ~x[3]) & (~x[2] | x[4]);
  f = g | h;
end
endmodule

```



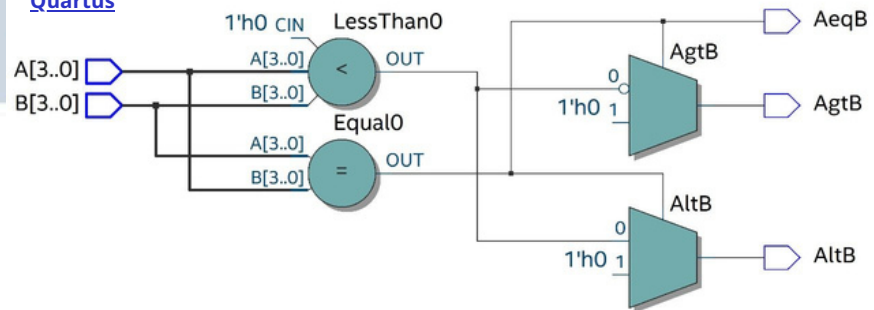
Task #3.3

```

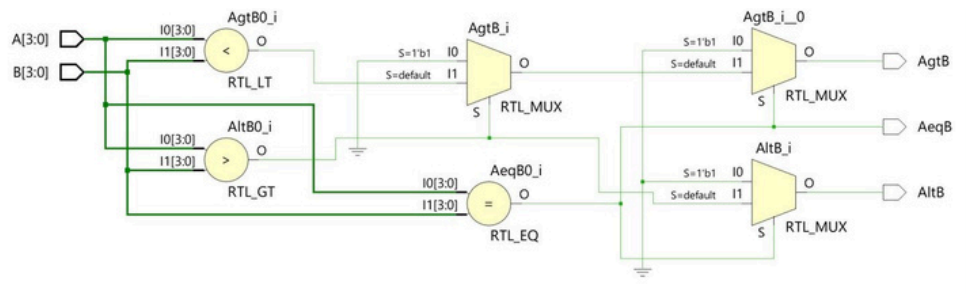
module task03
# (parameter n=4) (
input logic [n-1:0] A ,
input logic [n-1:0] B,
output logic AeqB,
output logic AltB,
output logic AgtB
);
always_comb begin
if (A==B) begin
AeqB=1;
AltB=0;
AgtB=0;
end
else if (A<B) begin
AeqB=0;
AltB=1;
AgtB=0;
end
else if (A>B) begin
AeqB=0;
AltB=0;
AgtB=1;
end
else begin
AeqB=1'bx;
AltB=1'bx;
AgtB= 1'bx;
end
end
endmodule

```

[Quartus](#)



[Vivado](#)



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task3.1	✓	✓	✓	✓	✓
Task3.2	✓	✓	✓	✓	✓
Task3.3	✓	✓	✓	✓	✓

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